FinResearch AI

Multi-Agent Financial Research System

THE PROBLEM

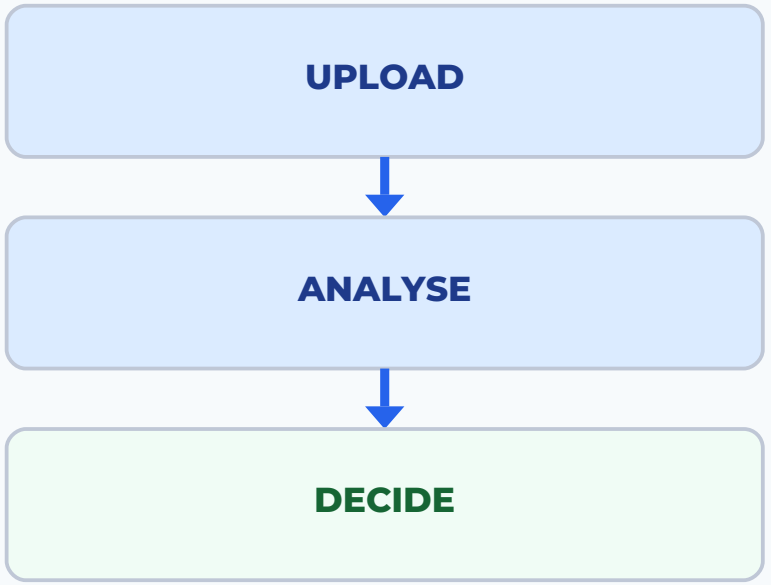
- Annual reports run 100+ pages and take hours to read and analyse by hand.
- AI chat tools invent numbers and show no sources for their claims.
- Teams cannot trust financial answers that they cannot verify.



How We Solve It

The solution

- You upload a filing, and eight specialised agents analyse it end to end.
- We extract 26 financial metrics and flag every risk with its exact source quote.
- Ask questions, compare companies, and generate a full research report in seconds.
- Every number on screen carries its evidence line from the original filing.





Why Multi-Agent?

Our approach

- One big AI prompt is a black box -- when it fails, everything fails with it.
- We split the job into eight focused agents, each with its own fallback path.
- Numbers are computed by reliable code; language is handled by AI.
- The result is a system that is testable, auditable, and explainable.

[x] Single LLM

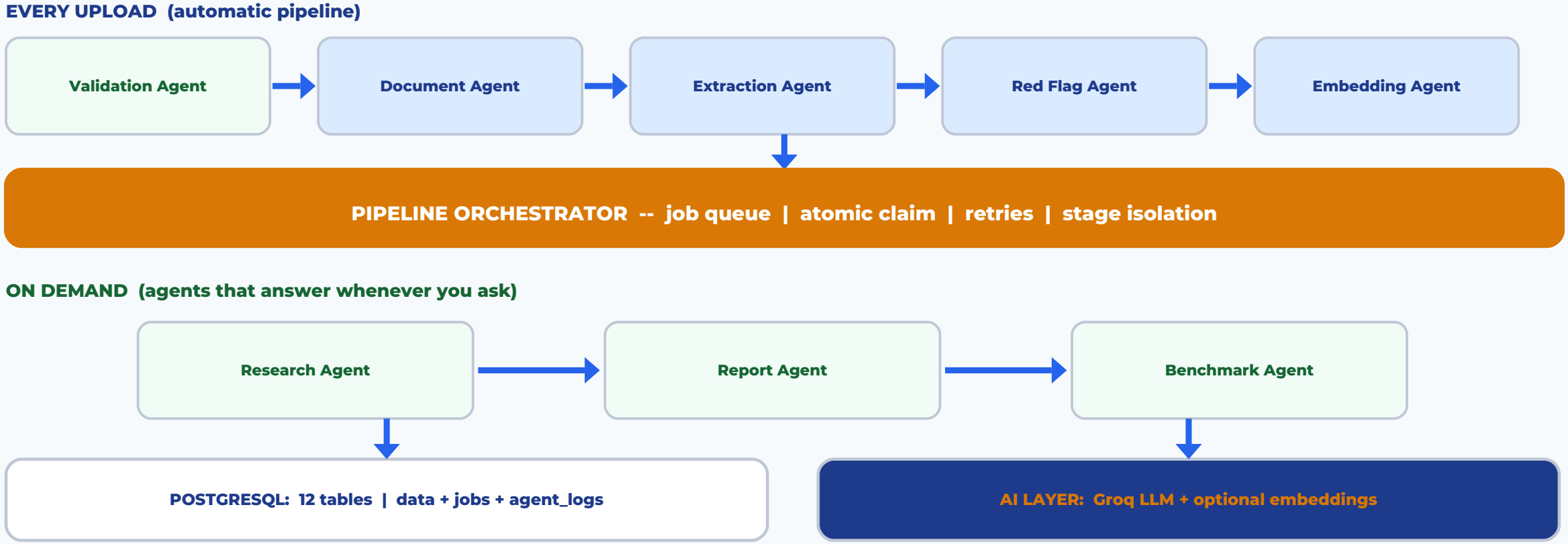
It hallucinates numbers, cites nothing, and one failure kills the whole answer.

[v] Multi-Agent

Verified numbers, cited answers, isolated failures -- one stage can fail while the rest continue.

How Agents Interact

Architecture -- all 8 agents + orchestrator



Orchestrator runs every stage; each stage writes to Postgres. If one stage fails, the others keep running -- failures stay isolated.

The Agents (1/2)

Automatic on every upload

Validation	Checks that the upload is truly a financial document -- resumes, invoices, and novels are rejected by name.
Document	Splits the report into searchable chunks and builds a full-text index of the document.
Extraction	Reads the report and extracts 26 financial metrics, each with its source evidence line.
Red Flag	Scans the filing language for risk signals like going concern and debt, always with exact quotes.
Embedding	Turns text into vectors for semantic search -- optional and works with any embeddings provider.

The Agents (2/2)

On demand + the controller

Research	Answers your questions with [S1]-style citations that you can click and verify against the filing.
Report	Builds the full research report -- tables generated by code, narrative written by AI from real data.
Benchmark	Compares companies and periods side by side, safely across fiscal-year gaps.
Orchestrator	Manages the job queue, retries failures automatically, and keeps every stage isolated.

Each agent does one job well, has its own tests, and has its own fallback.

Measuring The Metrics

Extraction and quality

- We measure 26 metrics per document: growth, margins, ratios, and cash flow.
- A regex label parser reads statements like 'Total revenue: $383,285 million' exactly as written.
- Every metric stores the source line it came from -- nothing is ever guessed.
- Quality gate: every AI answer must match the source text before it is shown.

```
Total revenue:  $383,285 million

Evidence: the exact filing line is saved with the metric
```

Verified by code,

not by AI



Built To Be Reliable

Graceful degradation

- If the AI provider is offline, deterministic rule-based modes take over automatically.
- If embeddings are disabled, keyword search keeps retrieval working for every query.
- If one stage crashes, the pipeline marks it partial and continues with the rest.
- Jobs run in the background with automatic retries and stale-lock recovery.
- Security: hashed passwords, scoped API routes, and validated uploads.

AI offline	->	Rule-based mode
Bad PDF	->	Clear, exact error
Stage crash	->	Status: partial



The System Today

Built and deployed

8	26	20	12	$0
AGENTS	METRICS	APIs	TABLES	COST

- **Eight agents, twenty APIs, and twelve database tables -- live and tested end to end.**
- **The full loop: upload a filing, receive risk flags, ask research questions, and export the report.**
- **Built entirely on free tiers -- zero infrastructure cost to run.**



Why It Matters

- Trustworthy AI: every claim is cited and every number is sourced.
- Multi-agent design makes the system robust by construction, not by luck.
- Built, tested, and deployed -- and it costs nothing to run.

Thank You

Questions?